

Abstract

Nearly half of the insect species in the world are projected to become extinct over the next few decades and is thought to occur in most environments globally. Climate change is one of the major driving factors behind insect decline. With the progress of global climate change, the direct effects of increasing temperatures, and the indirect effects of warming temperatures on factors such as soil and vegetation characteristics, are all potential drivers that could explain any changes in insect populations in their respective environments. There is evidence that changing environmental factors can cause suitable habitats for different fauna and flora to shift across landscapes and result in shifting of both the placement and extent of their distribution ranges. This can occur due to novel competition, exceeding critical physiological limits, and availability of food and habitats. Global insect decline is a pressing concern because insects are an ecologically diverse group of organisms that provide a wide variety of crucial ecosystem services to the environments they inhabit. They are also a useful indicator of the status of the other fauna that inhabit the same environments as them and can indicate that those respective environments might experience trophic or ecological collapse. The extent and potential environmental drivers of insect decline are especially understudied in the Southern Hemisphere. This relative lack of research when compared to other taxa has meant that, despite their ecological importance, insects are currently not widely being studied or tracked for conservation strategies in key environments. Therefore, the extent of insect decline in South Africa has not been well-documented. Beetles are the most species rich family of insects globally and exhibit great levels of ecological diversity. They provide crucial ecosystem services such as seed dispersal, decomposition, pest control, acting as prey for other species, and vegetation cover management. The Cape Floristic Region (CFR) is one of the most diverse floral kingdoms globally. The CFR is expected to contract in its extent as climate change progresses, meaning that the insects that rely on specific vegetation present in those environments will have their habitable ranges also contract in upcoming years, potentially resulting in accelerated decline and the ecological consequences resulting from that decline. A useful tool in allowing for effective observational studies on the potential effects of climate change on populations, are altitudinal gradients. Mountains especially provide a spatially continuous gradient of rapidly changing environmental conditions across their altitudes, allowing for observations of how populations of certain species assemble themselves throughout those changing environments. Analysing these different distributions can allow for the environmental preferences of different species to be extrapolated and can therefore be used as a proxy for how those species could respond to climate change. In this study, we aimed to assess changes in ground-dwelling beetle assemblages across an altitudinal gradient over a 20-year period. The historical study was performed in 2002, and repeated in October 2022, March 2023, and October 2023 by following the same methodology and locations as previously used. Tenebrionidae, Carabidae, and Cicindelidae beetles were sampled and identified as taxonomically far as possible. For the historical comparisons, three beetle species were compared over time, including *Anthia decemguttata* (Carabidae), *Stenocara dentata* (Tenebrionidae), and *Zophosis gracilicornis* (Tenebrionidae). These are the three most abundant species in this environment and exhibit wide trophic diversity. Along with the beetle sampling, the temperatures in the environment were modelled using

meteorological data, soil samples were collected at every site to determine soil characteristics, vegetation ground cover and vegetation height were estimated, and the fire history of the study site was considered. There was a 0.19°C increase in baseline air temperatures, and a 0.84°C increase in ground temperatures over time. Plant litter had increased by 183% and live vegetation had increased by 27% across the 20 years. There was a 97% decline in abundance and an 88% decline in species diversity between the October and March 2023 season. When compared to the historical data, across all three of the focus species there was a 47% decline in abundance overall. Most of this decline in abundance was due to population decline for the *Zophosis gracilicornis*, which reduced by 66% in abundance between the 2002-2003 samples compared to the 2022-2023 samples. These changes were strongly related to a combination of environmental factors, with increasing air temperatures, ground temperatures, and vegetation, all contributing to a reduced abundance of the species in question. These observations suggest that insect decline is occurring to some extent in this environment. The true extent of its intensity as well as the number of species it affects can only be determined by sampling other beetle families and identifying the present samples further to match with historical data. Describing the specific biology and ecological relationships of the species in question can also allow for a better understanding of the environmental drivers behind the observed changes. These results indicate that more focus needs to be placed on the beetle populations present within this environment so that, if necessary, the relevant conservation interventions can be put in place.